

C4: Yingchuan parameter checkpoint $\lambda=0.4$, $\beta=0.002$

Sapelo array 48487700; five prespecified exploratory seeds. This is one of eight sequential checkpoints. All five jobs are terminal before this report; no next beta was launched for its construction.

Evidence boundary: measured CIFAR-100-C severity-5 trajectories and finite intervention responses. Numerical failure is a failed computation, not behavioral deterioration. A finite F response does not identify a closed Jacobian, spectrum or phase transition.

Complete five-seed status

Seed	Parent blocks	Extension blocks	Forks verified	H2O analyses	Coverage / reason
202	15	30	4/4	2/2	complete
303	15	30	4/4	2/2	complete
404	15	30	4/4	2/2	complete
505	15	30	4/4	2/2	complete
606	15	30	4/4	2/2	complete

A component is reported only after its own completion and verification. Parent, extension, fork and H2O gaps remain explicit; the raw run and failed attempts remain on Sapelo.

Exploratory readout: visit-3 minus visit-1 same-domain CE is positive in 4/5 seeds (mean +0.166); classification error changes by -4.24 percentage points on average. At H1410, 5/10 correlated parent nodes have harmful closed acceptance and 5/10 have positive feedback interaction. The signs are mixed; these are descriptive, not a persistent-deterioration or stability claim.

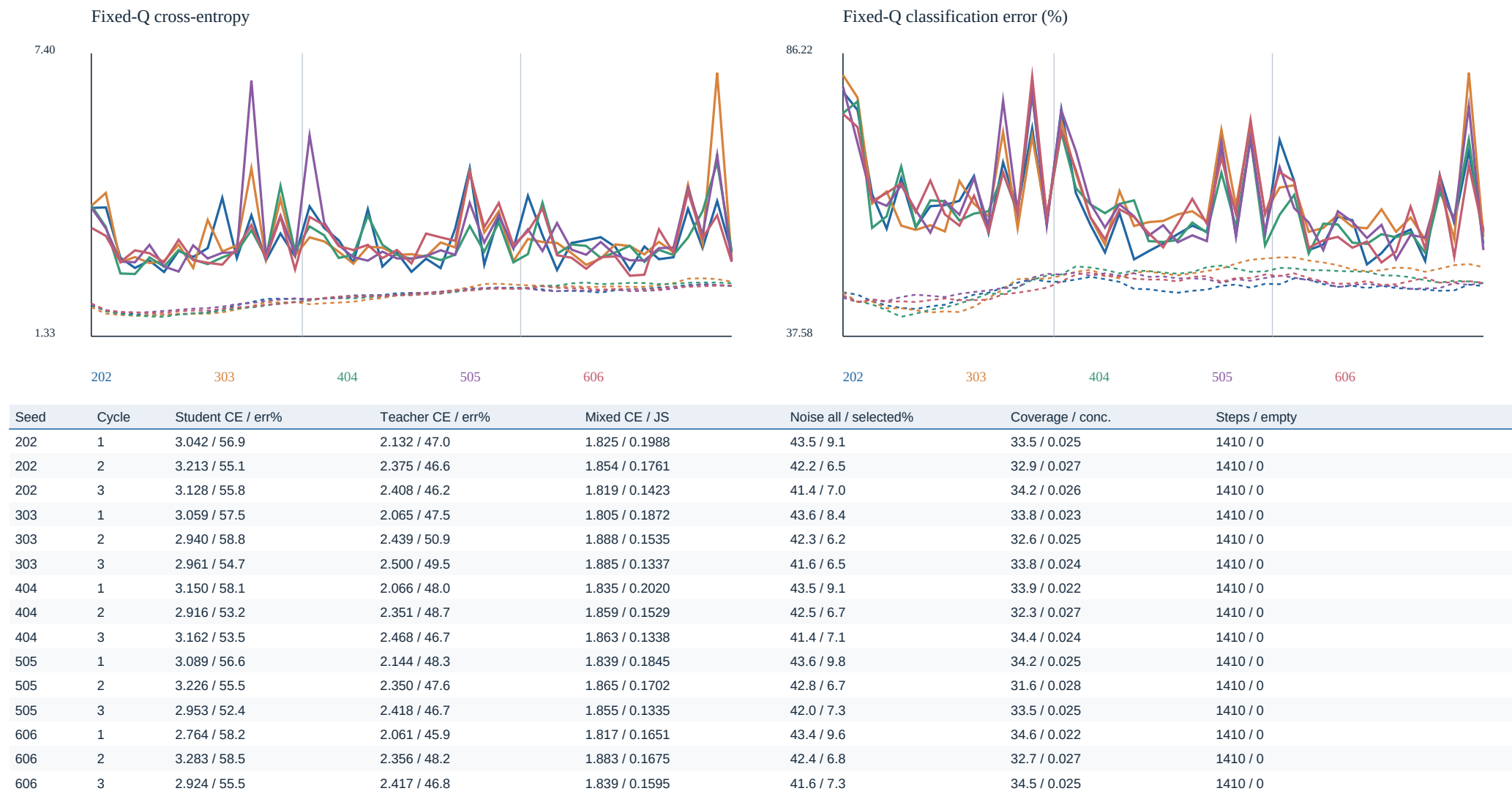
C4-1. Proxy versus labelled local update

Negative post-minus-pre CE means local improvement. Near zero means either absolute change is below 1e-4. Natural C4 executes updates without a separate accept/reject evaluator.

Seed	Probes	Both improve	Proxy down / local up	Proxy up / local down	Both worsen	Near zero
202	90	45	45	0	0	0
303	90	43	47	0	0	0
404	90	55	34	0	1	0
505	90	42	48	0	0	0
606	90	49	41	0	0	0

C4-2. Complete fixed-Q trajectories and pseudo-supervision

The fixed-Q panel uses the same 1,500 image-corruption evaluations at every block. Curves are unsmoothed. Dotted curves are teachers; solid curves are students. Missing/failed cells are absent, not assigned zero risk.



Cycle CSV retains teacher/student disagreement, mixed-target confidence, count-weighted pseudo-label statistics, selected-class concentration and elapsed block time. Selected noise is undefined when no examples were selected.

C4-3. Same-domain returns

Visit-3 minus visit-1 post-block student CE for each named corruption. Positive means worse on that domain; all domains and all terminal cells are retained. The companion block CSV includes pre/post CE and error for student and teacher at every visit.

Corruption	202	303	404	505	606
gaussian_noise	0.325	-0.937	-0.750	-0.818	-0.007
shot_noise	0.148	-0.831	1.118	-0.335	1.294
impulse_noise	0.057	0.715	0.403	0.655	0.149
defocus_blur	0.354	0.205	0.463	0.260	0.450
glass_blur	0.407	0.163	0.400	0.275	-0.470
motion_blur	0.704	0.332	0.440	0.454	0.503
zoom_blur	0.440	0.176	0.142	0.529	0.350
snow	-0.215	0.268	0.615	0.002	-0.325
frost	-0.113	-2.532	0.632	0.311	-0.203
fog	-0.729	0.099	0.126	0.472	0.327
brightness	0.312	0.280	0.422	0.357	0.219
contrast	0.199	-0.756	-0.595	0.094	-0.242
elastic_transform	0.412	0.337	0.792	0.260	0.337
pixelate	0.668	1.562	0.855	1.296	-0.474
jpeg_compression	0.204	-0.203	-0.494	0.080	0.022

C4-4. Per-seed parameter summary

Seed	Mean post CE / err%	Mean return dCE / dErr pp	Cycle-3 CE / err%	Coverage%	Updates	Block cost min
202	2.852 / 54.1	0.212 / -3.30	2.931 / 52.7	33.5	4230	15.2
303	2.862 / 54.0	-0.075 / -5.03	2.930 / 52.3	33.4	4230	15.2
404	2.805 / 53.8	0.305 / -4.04	2.967 / 52.0	33.5	4230	15.4
505	2.885 / 54.9	0.260 / -3.34	2.982 / 53.1	33.1	4230	48.6
606	2.818 / 54.1	0.129 / -5.49	2.815 / 51.1	33.9	4230	50.8

C4-5. Matched accept/reject and closed/replay forks

The first nonempty update after preselected blocks 5 and 15 is the focal candidate. Positive accept-minus-reject CE is worse. $A_{\text{closed}} = \text{accept_closed} - \text{reject_closed}$; $A_{\text{open}} = \text{accept_replay} - \text{reject_replay}$; $I_{\text{feedback}} = A_{\text{closed}} - A_{\text{open}}$. The H94 and H1410 forks share their initial state and schedule; their latest common scored checkpoint, H93, agrees exactly. H94 is therefore a correlated branch, not an independent replicate. The H1410 run does not separately score H94.

Seed	Parent	Immediate Q dCE / dErr pp	Immediate domain dCE / dErr pp
202	5	0.069 / 1.93	0.081 / 2.07
202	15	-0.117 / -0.20	-0.153 / -0.60
303	5	0.076 / 1.40	0.118 / 1.60
303	15	-0.025 / -0.93	-0.063 / -0.73
404	5	-0.064 / -1.33	-0.117 / -0.60
404	15	0.125 / 0.67	0.481 / 4.27
505	5	0.066 / -0.33	0.033 / -0.87
505	15	0.074 / 0.47	0.384 / 1.67
606	5	-0.140 / -2.07	-0.219 / -3.40
606	15	0.015 / -0.60	0.115 / 0.40

H94 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_{closed}	A_{open}	I_{feedback}
202	5	2.555	2.555	2.723	2.698	0.169	0.143	0.026
202	15	4.923	4.923	4.213	3.471	-0.711	-1.453	0.742
303	5	2.644	2.644	2.807	2.725	0.163	0.081	0.082
303	15	3.565	3.565	3.420	3.440	-0.145	-0.125	-0.020
404	5	2.887	2.887	2.873	3.114	-0.014	0.227	-0.241
404	15	3.816	3.816	3.616	3.831	-0.201	0.014	-0.215
505	5	3.212	3.212	2.775	2.712	-0.437	-0.500	0.063
505	15	3.876	3.876	5.891	4.547	2.016	0.671	1.344
606	5	3.303	3.303	2.874	3.167	-0.429	-0.136	-0.292
606	15	3.445	3.445	3.863	3.790	0.418	0.345	0.073

H1409 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	3.163	3.163	4.055	3.232	0.892	0.070	0.822
202	15	2.977	2.977	3.213	3.010	0.236	0.033	0.203
303	5	3.384	3.384	3.250	3.327	-0.134	-0.057	-0.077
303	15	3.268	3.268	2.940	3.130	-0.327	-0.137	-0.190
404	5	3.737	3.737	3.917	3.040	0.180	-0.697	0.877
404	15	2.945	2.945	2.916	3.378	-0.029	0.433	-0.462
505	5	3.073	3.073	2.952	3.278	-0.120	0.205	-0.326
505	15	3.309	3.309	3.226	3.478	-0.084	0.168	-0.252
606	5	2.955	2.955	3.292	3.000	0.336	0.044	0.292
606	15	3.054	3.054	3.283	2.845	0.230	-0.208	0.438

H1410 fixed-Q CE contrasts

Seed	Parent	R closed	R replay	A closed	A replay	A_closed	A_open	I_feedback
202	5	3.127	3.127	4.071	3.268	0.944	0.141	0.803
202	15	2.942	2.942	3.224	2.941	0.282	-0.001	0.283
303	5	3.481	3.481	3.278	3.461	-0.204	-0.020	-0.184
303	15	3.291	3.291	2.899	3.125	-0.392	-0.166	-0.225
404	5	3.717	3.717	3.913	3.072	0.196	-0.645	0.841
404	15	2.913	2.913	2.906	3.424	-0.008	0.511	-0.518
505	5	3.081	3.081	2.978	3.266	-0.103	0.186	-0.289
505	15	3.370	3.370	3.129	3.396	-0.242	0.025	-0.267
606	5	2.946	2.946	3.381	3.015	0.435	0.069	0.366
606	15	3.127	3.127	3.296	2.846	0.168	-0.281	0.450

C4-6. Diagnostic state, finite F and local identifiability

x=(student fixed-Q CE, mixed-target fixed-Q CE, student/teacher JS, mixed confidence). F is the mean H20 future x over three base tapes at each observed parent; it is a conditional response, not a closed transition law. The block CSV stores every observed x_t .

Seed	Parent	x0	F_H20(x0)	Tape SD	Finite	J status
202	5	2.953 / 1.787 / 0.262 / 0.654	2.824 / 1.782 / 0.196 / 0.652	0.107 / 0.000 / 0.013 / 0.000	132/132	not identified
202	15	3.042 / 1.825 / 0.199 / 0.636	3.541 / 1.823 / 0.287 / 0.635	0.211 / 0.001 / 0.028 / 0.000	132/132	not identified
303	5	2.900 / 1.772 / 0.202 / 0.655	2.794 / 1.765 / 0.195 / 0.654	0.439 / 0.002 / 0.042 / 0.000	132/132	not identified
303	15	3.059 / 1.805 / 0.187 / 0.637	3.571 / 1.804 / 0.275 / 0.637	0.475 / 0.000 / 0.051 / 0.000	132/132	not identified
404	5	3.022 / 1.771 / 0.260 / 0.656	2.660 / 1.761 / 0.195 / 0.655	0.097 / 0.001 / 0.014 / 0.000	132/132	not identified
404	15	3.150 / 1.835 / 0.202 / 0.634	3.621 / 1.834 / 0.286 / 0.634	0.427 / 0.001 / 0.029 / 0.000	132/132	not identified
505	5	3.291 / 1.816 / 0.218 / 0.655	2.767 / 1.813 / 0.193 / 0.654	0.170 / 0.000 / 0.014 / 0.000	132/132	not identified
505	15	3.089 / 1.839 / 0.185 / 0.628	3.742 / 1.841 / 0.271 / 0.628	0.100 / 0.001 / 0.046 / 0.000	132/132	not identified
606	5	3.111 / 1.805 / 0.246 / 0.658	2.634 / 1.798 / 0.173 / 0.657	0.056 / 0.001 / 0.010 / 0.000	132/132	not identified
606	15	2.764 / 1.817 / 0.165 / 0.640	3.286 / 1.820 / 0.225 / 0.640	0.362 / 0.001 / 0.031 / 0.000	132/132	not identified

Perturbation, conditioning and held-out diagnostics

Seed	Parent	Median G	Rank	Condition	Scale diff	Held-out median	Closure	Reason
202	5	16.7	4	4.1	0.86	41.75	falsified	not established
202	15	25.7	4	3.7	0.56	132.09	falsified	not established
303	5	20.8	4	3.5	0.70	46.24	falsified	not established
303	15	47.5	4	2.9	1.59	187.17	falsified	not established
404	5	12.0	4	4.5	0.49	31.21	falsified	not established
404	15	87.9	4	3.4	0.28	397.38	falsified	not established
505	5	24.3	4	7.3	1.02	45.86	falsified	not established
505	15	66.9	4	1.9	1.14	453.75	falsified	not established
606	5	34.6	4	3.6	1.63	110.79	falsified	not established
606	15	31.3	4	5.6	1.20	139.48	falsified	not established

Independent student/teacher perturbations, input rank/condition, two scales, held-out directions and momentum closure checks are retained in the compact analysis files. No eigenvalue or phase boundary is claimed without a closed and predictive state law.

C4-7. Bounded parameter-map reading

This checkpoint isolates one pair $\lambda=0.4$, $\beta=0.002$ with five exploratory seeds. Seeds are reused across later pairs for paired comparison; they are not independent cross-parameter replications. Outcomes at this pair alone cannot locate a stability boundary. Numerical failures and improvement/null findings remain in the reported denominator.

Observed coverage: 5/5 complete 45-block trajectories, 10 verified H1410 parent nodes, 5 positive A_{closed} and 5 positive I_{feedback} . Parent nodes within one seed are correlated. A positive endpoint or interaction alone does not establish sustained recursive deterioration.

Coverage ledger and missingness

C4 item	Present evidence	Missing / limitation
C4-1	450 completed local probes	Unavailable after failed/incomplete trajectories
C4-2	5 full trajectories and cycle process records	Failed/partial cells not extrapolated
C4-3	75 same-domain return comparisons	Only complete first/third visits
C4-4	5 per-seed parameter summaries	One pair cannot characterize a sweep
C4-5	30 fork horizons	Unavailable parent nodes explicitly excluded
C4-6	10 H20 parent analyses	J/spectrum only if closure and held-out criteria pass
C4-7	One-pair status/sign description	No identified region or boundary

Provenance: Sapelo 48487700, compact evidence MANIFEST.json and AUDIT.json. Raw checkpoints, logits, step tapes and all failures remain under /scratch/yz54720/RSI2RSD/C4/sweep_yingchuan_20260923/results/I04_b0002/. The reserved confirmation panel was not read.